

Part 2: Quantitative data analysis

Quantitative Session of the 2026 HUSOE ELPS Methods and Data Institute

Nathan Alexander, PhD

Table of contents

0.1	Case Study: HBCUs and Educational Geography	1
0.1.1	Step 1: Load Packages	1
0.1.2	Step 2: Load the data	2
0.1.3	Step 3: Clean the data	3
0.1.4	Step 4: Basic Exploration	3
0.1.5	Step 5: Historical Analysis	3
0.1.6	Step 6: Geographic Distribution	4
0.1.7	Step 7: Mapping	5
0.2	Case Study: Data on State School Districts	6
0.2.1	Step 1: Load the libraries	6
0.2.2	Step 2: Load the data	6
0.2.3	Step 3: Clean the data	8
0.2.4	Step 4: Feature engineering	8
0.3	Recommended readings	10

0.1 Case Study: HBCUs and Educational Geography

In this section, we analyze data on Historically Black Colleges and Universities (HBCUs) to explore patterns in location, institutional type, and founding history.

0.1.1 Step 1: Load Packages

```
#install.packages("tidyverse")
#install.packages("lubridate")
library(tidyverse)
library(lubridate)
```

We'll start with a guided example using data on Historically Black Colleges and Universities.

0.1.2 Step 2: Load the data

```
library(tidyverse)
hbcu <- read_csv("https://raw.githubusercontent.com/quant-shop/intro-comp-educ-soc/refs/heads/main/data/hbcu.csv")

# View the first few rows
head(hbcu)
```

```
# A tibble: 6 x 7
  name          city      state founded   lat   lon type
  <chr>         <chr>    <chr>   <dbl> <dbl> <dbl> <chr>
1 Alabama A&M University Normal    AL      1875  34.8 -86.6 Public, 4 Year
2 Alabama State University Montgomery AL      1867  32.4 -86.3 Public, 4 Year
3 Albany State University Albany     GA      1903  31.6 -84.2 Public, 4 Year
4 Alcorn State University Lorman    MS      1871  31.9 -91.1 Public, 4 Year
5 Allen University      Columbia SC      1870  34.0 -81.0 Private, 4 Year
6 American Baptist College Nashville TN       1924  36.2 -86.8 Private, 4 Year
```

```
# Check structure
glimpse(hbcu)
```

```
Rows: 102
Columns: 7
$ name      <chr> "Alabama A&M University", "Alabama State University", "Albany ~
$ city      <chr> "Normal", "Montgomery", "Albany", "Lorman", "Columbia", "Nashv~
$ state     <chr> "AL", "AL", "GA", "MS", "SC", "TN", "AR", "SC", "NC", "FL", "A~
$ founded   <dbl> 1875, 1867, 1903, 1871, 1870, 1924, 1884, 1870, 1873, 1904, 19~
$ lat       <dbl> 34.7834, 32.3643, 31.5785, 31.8769, 34.0298, 36.1659, 34.7465,~
$ lon       <dbl> -86.5683, -86.2952, -84.1543, -91.1458, -81.0115, -86.7844, -9~
$ type      <chr> "Public, 4 Year", "Public, 4 Year", "Public, 4 Year", "Public,~
```

```
# Summary statistics
summary(hbcu)
```

name	city	state	founded
Length:102	Length:102	Length:102	Min. :1837
Class :character	Class :character	Class :character	1st Qu.:1870
Mode :character	Mode :character	Mode :character	Median :1886
			Mean :1895
			3rd Qu.:1905
			Max. :1988

lat	lon	type
Min. :18.34	Min. :-98.50	Length:102
1st Qu.:32.48	1st Qu.: -90.13	Class :character

```

Median :34.02   Median :-84.64   Mode  :character
Mean   :34.31   Mean    :-85.13
3rd Qu.:36.17   3rd Qu.: -80.78
Max.    :39.93   Max.     :-64.96

```

0.1.3 Step 3: Clean the data

```

hbcu <- hbcu %>%
  mutate(
    founded = as.numeric(founded),
    type = as.factor(type),
    state = as.factor(state)
  )

```

0.1.4 Step 4: Basic Exploration

How many institutions are in the data set?

```
nrow(hbcu)
```

```
[1] 102
```

What is the distribution by HBCU type?

```

hbcu %>%
  count(type) %>%
  arrange(desc(n))

```

```

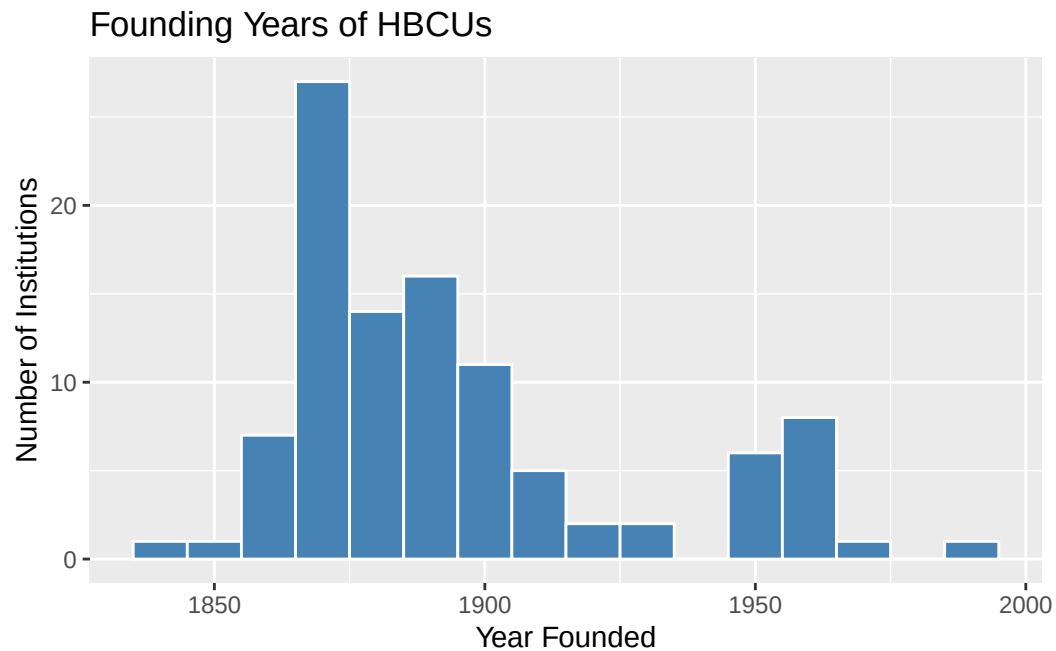
# A tibble: 5 x 2
  type                n
  <fct>             <int>
1 Private, 4 Year      45
2 Public, 4 Year       40
3 Public, 2 Year       11
4 Private, Specialized  4
5 Private, 2 Year       2

```

0.1.5 Step 5: Historical Analysis

When were the HBCUs founded?

```
hbcu %>%
  ggplot(aes(x = founded)) +
  geom_histogram(binwidth = 10, fill = "steelblue", color = "white") +
  labs(
    title = "Founding Years of HBCUs",
    x = "Year Founded",
    y = "Number of Institutions"
  )
```



0.1.6 Step 6: Geographic Distribution

HBCUs by state

```
hbcu %>%
  count(state, sort = TRUE)
```

```
# A tibble: 21 x 2
  state      n
  <fct> <int>
1 AL      14
2 GA      10
3 NC      10
4 TX       9
5 SC       8
6 MS       7
7 LA       6
8 TN       6
```

```

9 AR      4
10 FL     4
# i 11 more rows

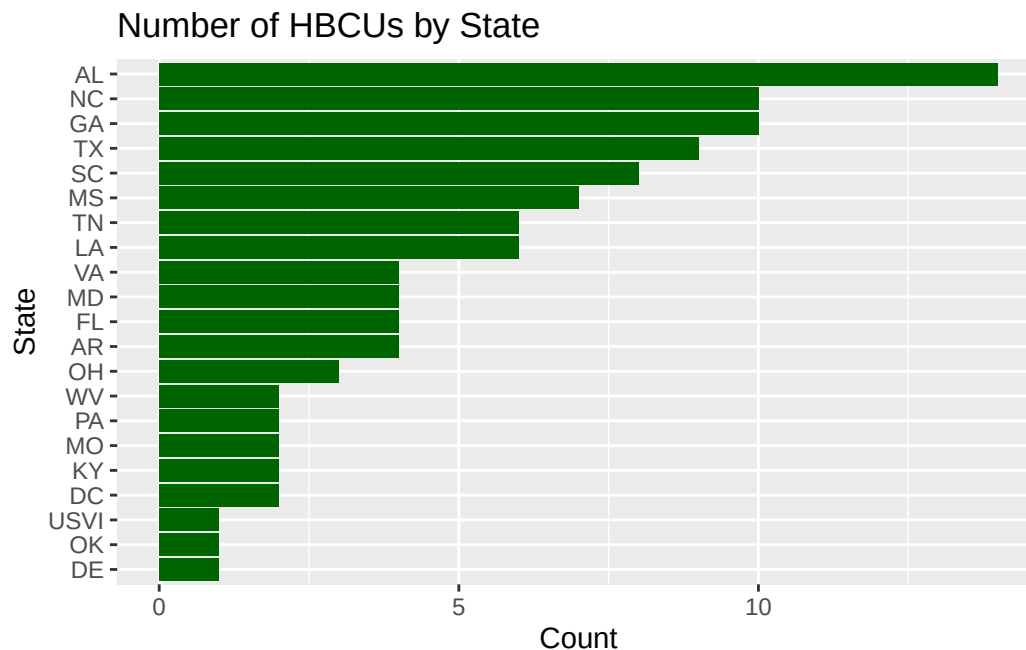
```

Basic visualization of HBCUs by state

```

hbcu %>%
  count(state) %>%
  ggplot(aes(x = reorder(state, n), y = n)) +
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(
    title = "Number of HBCUs by State",
    x = "State",
    y = "Count"
  )

```



0.1.7 Step 7: Mapping

```

#install.packages("maps", repos = "http://cran.us.r-project.org")
library(maps)

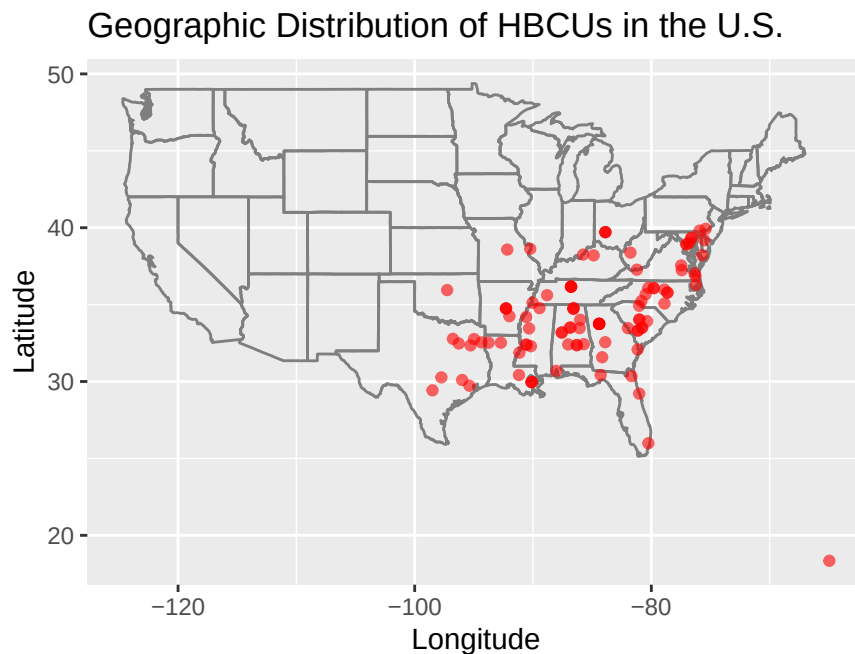
```

```

hbcu %>%
  ggplot(aes(x = lon, y = lat)) +
  borders("state") +
  geom_point(color = "red", alpha = 0.6) +

```

```
coord_fixed(1.3) +
labs(
  title = "Geographic Distribution of HBCUs in the U.S.",
  x = "Longitude",
  y = "Latitude"
)
```



0.2 Case Study: Data on State School Districts

0.2.1 Step 1: Load the libraries

```
library(dplyr)
library(readr)
library(stringr)
library(janitor)
```

0.2.2 Step 2: Load the data

```
district <- read.csv("https://raw.githubusercontent.com/quant-shop/intro-comp-educ-soc/refs/heads/main/data/state-district-data.csv")
# district <- read.csv("../data/state-district-data.csv")

str(district)
```

```

'data.frame':  57 obs. of  36 variables:
 $ State.Name                : chr  "ALABAMA" "ALASKA"
 $ State.Abbbr               : chr  "AL " "AK " "AS "
 $ ANSI.FIPS.State.Code..State..Latest.available.year : int  1 2 60 4 5 59 6 8
 $ Total.Number.Operational.School.Districts..District..2023.24 : int  156 54 1 685 304
 $ Total.Number.Operational.Schools..Public.School..2023.24 : int  1524 494 29 2427
 $ Total.Number.Operational.Charter.Schools..Public.School..2023.24 : chr  "18" "31" "†" "57"
 $ Prekindergarten.Students..State..2023.24 : chr  "21924" "3320" "-"
" "20810" ...
 $ Kindergarten.Students..State..2023.24 : chr  "55881" "9257" "-"
" "72320" ...
 $ Grades.1.8.Students..State..2023.24 : chr  "447317" "78890"
" "654843" ...
 $ Grades.9.12.Students..State..2023.24 : chr  "223528" "39776"
" "369124" ...
 $ Grade.13.Students..State..2023.24 : chr  "†" "†" "†" "†"
 $ Ungraded.Students..State..2023.24 : chr  "†" "†" "†" "533"
 $ American.Indian.Alaska.Native.Students..State..2023.24 : chr  "5997" "28284" "-"
" "46878" ...
 $ Asian.or.Asian.Pacific.Islander.Students..State..2023.24 : chr  "11016" "6458" "-"
" "35975" ...
 $ Hispanic.Students..State..2023.24 : chr  "84196" "10076"
" "539087" ...
 $ Black.or.African.American.Students..State..2023.24 : chr  "236304" "2955"
" "64938" ...
 $ White.Students..State..2023.24 : chr  "381550" "61502"
" "378320" ...
 $ Nat..Hawaiian.or.Other.Pacific.Isl..Students..State..2023.24 : chr  "831" "4148" "-"
" "4017" ...
 $ Two.or.More.Races.Students..State..2023.24 : chr  "28756" "17820"
" "48415" ...
 $ Total.Enrollment..Exclude.AE..for.SY.2014.15.onward..State..2023.24 : chr  "748650" "131243"
" "1117630" ...
 $ Full.Time.Equivalent..FTE..Teachers..State..2023.24 : chr  "42857.92" "7221.5"
" "49260.47" ...
 $ Pupil.Teacher.Ratio..State..2023.24 : chr  "17.47" "18.17"
" "22.69" ...
 $ State.Name..State..2023.24 : chr  "ALABAMA" "ALASKA"
 $ Total.Number.of.Districts.with.Enrollment..District..2023.24 : int  150 54 0 664 260
 $ Total.Number.of.School.Districts..District..2023.24 : int  159 54 1 722 308
 $ Total.Number.of.Public.Schools..Public.School..2023.24 : int  1527 498 29 2557
 $ Male.Students..State..2023.24 : chr  "384500" "67646"
 $ Female.Students..State..2023.24 : chr  "364150" "63597"
 $ Black.or.African.American...male..State..2023.24 : chr  "120605" "1488"
" "33191" ...
 $ Black.or.African.American...female..State..2023.24 : chr  "115699" "1467"
" "31747" ...
 $ White...male..State..2023.24 : chr  "197384" "31786"

```

```

" "194292" ...
$ White...female..State..2023.24 : chr "184166" "29716"
" "184028" ...
$ Grade.12.Students...Black.or.African.American..State..2023.24 : chr "15413" "241" "-"
" "5876" ...
$ Secondary.Teachers..State..2023.24 : chr "20188.71" "3673."
" "15996.72" ...
$ Elementary.Teachers..State..2023.24 : chr "18395.46" "2985."
" "30478.09" ...
$ Full.Time.Equivalent..FTE..Staff..State..2023.24 : chr "79963.90" "16423"
" "109199.43" ...

```

0.2.3 Step 3: Clean the data

```

district_clean <- district %>%

# Fix missing/suppressed values
mutate(across(where(is.character), ~na_if(., "-"))) %>%
mutate(across(where(is.character), ~na_if(., "†"))) %>%
mutate(across(where(is.character), str_trim)) %>%

# Convert numeric columns safely
mutate(across(
  where(is.character),
  ~parse_number(.)
))

```

0.2.4 Step 4: Feature engineering

We can create variables that answer educational questions.

```
str(district_clean)
```

```

'data.frame': 57 obs. of 36 variables:
 $ State.Name : num NA NA NA NA NA NA
 ..- attr(*, "problems")= tibble [57 x 4] (S3: tbl_df/tbl/data.frame)
 .. ..$ row : int [1:57] 1 2 3 4 5 6 7 8 9 10 ...
 .. ..$ col : int [1:57] NA NA NA NA NA NA NA NA NA NA ...
 .. ..$ expected: chr [1:57] "a number" "a number" "a number" "a number" ...
 .. ..$ actual : chr [1:57] "ALABAMA" "ALASKA" "AMERICAN SAMOA" "ARIZONA" ...
 $ State.Abbbr : num NA NA NA NA NA NA
 ..- attr(*, "problems")= tibble [57 x 4] (S3: tbl_df/tbl/data.frame)
 .. ..$ row : int [1:57] 1 2 3 4 5 6 7 8 9 10 ...
 .. ..$ col : int [1:57] NA NA NA NA NA NA NA NA NA NA ...
 .. ..$ expected: chr [1:57] "a number" "a number" "a number" "a number" ...

```



```

.. ..$ actual : chr [1:57] "AL" "AK" "AS" "AZ" ...
$ ANSI.FIPS.State.Code..State..Latest.available.year : int 1 2 60 4 5 59 6 8
$ Total.Number.Operational.School.Districts..District..2023.24 : int 156 54 1 685 304
$ Total.Number.Operational.Schools..Public.School..2023.24 : int 1524 494 29 2427
$ Total.Number.Operational.Charter.Schools..Public.School..2023.24 : num 18 31 NA 571 105
$ Prekindergarten.Students..State..2023.24 : num 21924 3320 NA 208
$ Kindergarten.Students..State..2023.24 : num 55881 9257 NA 723
$ Grades.1.8.Students..State..2023.24 : num 447317 78890 NA 6
$ Grades.9.12.Students..State..2023.24 : num 223528 39776 NA 3
$ Grade.13.Students..State..2023.24 : num NA NA NA NA NA NA
$ Ungraded.Students..State..2023.24 : num NA NA NA 533 174
$ American.Indian.Alaska.Native.Students..State..2023.24 : num 5997 28284 NA 468
$ Asian.or.Asian.Pacific.Islander.Students..State..2023.24 : num 11016 6458 NA 359
$ Hispanic.Students..State..2023.24 : num 84196 10076 NA 53
$ Black.or.African.American.Students..State..2023.24 : num 236304 2955 NA 64
$ White.Students..State..2023.24 : num 381550 61502 NA 3
$ Nat..Hawaiian.or.Other.Pacific.Isl..Students..State..2023.24 : num 831 4148 NA 4017
$ Two.or.More.Races.Students..State..2023.24 : num 28756 17820 NA 48
$ Total.Enrollment..Exclude.AE..for.SY.2014.15.onward..State..2023.24 : num 748650 131243 NA
$ Full.Time.Equivalent..FTE..Teachers..State..2023.24 : num 42858 7221 NA 492
$ Pupil.Teacher.Ratio..State..2023.24 : num 17.5 18.2 NA 22.7
$ State.Name..State..2023.24 : num NA NA NA NA NA NA
..- attr(*, "problems")= tibble [57 x 4] (S3: tbl_df/tbl/data.frame)
.. ..$ row : int [1:57] 1 2 3 4 5 6 7 8 9 10 ...
.. ..$ col : int [1:57] NA NA NA NA NA NA NA NA NA NA ...
.. ..$ expected: chr [1:57] "a number" "a number" "a number" "a number" ...
.. ..$ actual : chr [1:57] "ALABAMA" "ALASKA" "AMERICAN SAMOA" "ARIZONA" ...
$ Total.Number.of.Districts.with.Enrollment..District..2023.24 : int 150 54 0 664 260
$ Total.Number.of.School.Districts..District..2023.24 : int 159 54 1 722 308
$ Total.Number.of.Public.Schools..Public.School..2023.24 : int 1527 498 29 2557
$ Male.Students..State..2023.24 : num 384500 67646 NA 5
$ Female.Students..State..2023.24 : num 364150 63597 NA 5
$ Black.or.African.American...male..State..2023.24 : num 120605 1488 NA 33
$ Black.or.African.American...female..State..2023.24 : num 115699 1467 NA 31
$ White...male..State..2023.24 : num 197384 31786 NA 1
$ White...female..State..2023.24 : num 184166 29716 NA 1
$ Grade.12.Students...Black.or.African.American..State..2023.24 : num 15413 241 NA 5876
$ Secondary.Teachers..State..2023.24 : num 20189 3673 NA 159
$ Elementary.Teachers..State..2023.24 : num 18395 2985 NA 304
$ Full.Time.Equivalent..FTE..Staff..State..2023.24 : num 79964 16423 NA 10
..- attr(*, "problems")= tibble [1 x 4] (S3: tbl_df/tbl/data.frame)
.. ..$ row : int 51
.. ..$ col : int NA
.. ..$ expected: chr "a number"
.. ..$ actual : chr "†"

```

```
# what kinds of questions might you answer with this data set?
```

0.3 Recommended readings

Knafllic, C. N. (2015). *Storytelling with data: A data visualization guide for business professionals*. Wiley.